

4th April 2023, Slot: 8:30 AM – 09:30 AM

Course Code: CS AI2002	Course Name: Artificial Intelligence
Instructor Name:	Dr Fahad / Saeeda Kanwal / Dr.Waheed Ahmed/ Dr Muhammad Farrukh Shahid
Student Roll No:	Section No:

- Return the question paper.
- Read each question completely before answering it. There are **4 questions and 2 pages**.
- In case of any ambiguity, you may make assumption. But your assumption should not contradict with any statement in the question paper.
- All the answers must be solved according to the sequence given in the question paper.
- Be specific, to the point while coding, logic should be properly commented, and illustrate with diagram where necessary.

Time: 60 minutes.

Max Marks: 50 points

Question no1

CLO3 Marks [15]

Consider a robot who assigns task to the worker based on certain constraints. Robot has got a list of tasks that must be done, a list of workers who can do each task, and a list of times at which each task must be done. Followings are constrained by the fact that each worker can only assigned to one task at a time. Below table provide detail of when these tasks to done and tasks worker can do.

Tasks (Must be done)	Work can be done by	Constraints
T1: from Nov 1 to Nov 4. One worker for all 4 days.	Worker 1 = {T3, T4, T5}	Every task must be started on its assigned start date.
T2: from Nov 2 to Nov 6. One worker for all 5 days.		Every task must continue for the length of time stated above for that task.
T3: from Nov 3 to Nov 8. One worker for all 6 days.	Worker 2 = {T1, T2, T4, T5}	Every task must be assigned to a worker who can do that task.
T4: from Nov 7 to Nov 18. One worker for all 12 days.	Worker 3 = {T1, T3, T5}	Once a worker is assigned to a task that worker must work only on that task until finished.
T5: from Nov 10 to Nov 11. One worker for all 2 days.		Only one worker may be assigned to each task.

Formulate this problem as a CSP problem in which there is one variable for each task, and so you have five variables, T1-T5. The domain of each variable is the set of workers who can do that task.

- Enumerate the domains of each variable. (5 points, 1 point each)
- Draw the constraint graph for variables T1-T5. (2 points)
- Assume that T2 is already assigned to its only domain value. What variables might be assigned by the Minimum Remaining Values (MRV) heuristic? (3 points)
- Show the domains of the variables after running arc-consistency on this initial graph. (5 points)

Question no2**CLO3 Marks [15]**

A: Suppose that three companies, **Company X**, **Company Y**, and **Company Z**, are competing in the market for a new product. Each company can choose to invest in research and development (R&D) or not invest in R&D. The offer is:

- A. If all companies invest in R&D, they will all develop a high-quality product and receive a benefit of \$10 million each.
- B. If two companies invest in R&D, they will all develop a high-quality product and receive a benefit of \$10 million each and the other companies receive nothing.
- C. If only one company invests in R&D, that company will develop a superior product and receive a benefit of \$20 million while the other companies receive nothing.
- D. If no company invests in R&D, they will all develop a mediocre product and receive a benefit of \$5 million each.

Now choose the best strategy for Company X using Min-Max concept and provide reasoning for your choice. Define properly how you generated utility values based on which the result/ strategy has been figured.

B: A game is played between Max, and Min. Draw a tree considering Max as a first player and branching factor is 2. Given terminal values below, show backed up values, best decision available at the root and branches that will get pruned (explain why they are getting pruned if any)?

2,3,1,3,5,4,13,1, -13, -9, -25, -27,7,8,9,10

Question no3**CLO3 Marks [10]**

Given the above configuration of a limited version of the Wumpus world with only Wumpus and its accompanying stench. $W_{x,y}$ is true if there is a Wumpus in $[x,y]$. $S_{x,y}$ is true if agent perceives a stench in $[x,y]$. Now do the following:

i. Create a Knowledge Base using predicate logic given the above configuration. Observations should be named as obs_1, obs_2....obs_n.

ii. Suppose $a1 = \{\text{No Wumpus in } 2,2\}$

Prove whether the Knowledge-Base entails $a1$ or not, using enumeration of all states.

1,2	2,2	3,2
1,1	2,1	3,1
Agent	Stench	Wumpus

Question no4**CLO3 Marks [10]**

Explain the following in one or two statements.

- 1) What is meant by Zero-Sum game?
- 2) Define successor function in min-max game.
- 3) What is the significance of game tree?
- 4) What are the types of constraints available in the constraint satisfaction problem?
- 5) What is meant by back tracking search?
- 6) Explain the binary constraint satisfaction problem.
- 7) What is meant by Logic?
- 8) Differentiate between propositional logic and predicate logic.
- 9) Explain the knowledge base.
- 10) Describe the importance of game theory in the wireless communication network.

Have a wonderful day 😊